14.2 Using ConfigMaps and Secrets in Kubernetes

The 15-Factor methodology recommends keeping code, configuration, and credentials always separate. Kubernetes fully embraces that principle and defines two APIs to handle configuration and credentials independently: ConfigMaps and Secrets. This section will introduce this new configuration strategy, which is provided natively by Kubernetes.

Spring Boot provides native and flexible support for both ConfigMaps and Secrets. I'll show you how to work with ConfigMaps and their relationships with environment variables, which are still a valid configuration option in Kubernetes. You'll see that Secrets are not really secret, and you'll learn what to do to make them really so. Finally, I'll go through a few options for dealing with configuration changes and propagating them to applications.

Before moving on further, let's set the scene and start a local Kubernetes cluster. Go to your Polar Deployment project (polar-deployment), navigate to the kubernetes/ platform/development folder, and run the following command to start a minikube cluster and deploy the backing services used by Polar Bookshop:

```
$ ./create-cluster.sh
```

NOTE If you haven't followed along with the examples implemented in the previous chapters, you can refer to the repository accompanying the book (https://github.com/ThomasVitale/cloud-native-spring-in-action) and use the projects in Chapter14/14-begin as a starting point.

The command will take a few minutes to complete. When it's finished, you can verify that all the backing services are ready and available with the following command: